ENTERPRISE HRMS + HR SERVICES PLATFORM

PHASE 0.1 — PRODUCT DISCOVERY & MVP SCOPE

Prepared for: CodeApp Technologies

Document status: Initial discovery baseline; architecture choices provisionally evaluated, not all vendor decisions finalized.

Executive summary

Define a realistic first release for a solo developer while preserving the ability to grow into an enterprise-grade, multi-tenant HR operating system. The MVP focuses on Core HRMS, Leave & Attendance, Basic Recruitment, and Employee Self-Service. Payroll, marketplace, advanced analytics, and AI automation remain outside the initial MVP.

Product vision

Build a multi-tenant HR management SaaS platform for Indian small and medium-sized businesses to manage employee records, leave, attendance, basic recruitment, and employee self-service from one application.

Project dimension	Phase 0.1 baseline
Target customers	SMBs first, with enterprise scalability
Initial market	India
Development model	Solo developer working with ChatGPT
Product delivery	Browser-based SaaS assumed; validate in discovery
MVP modules	Core HRMS; Leave & Attendance; Basic Recruitment; Employee Self-Service
Architecture style	Multi-tenant SaaS; modular monolith; API-first

Source of truth: the uploaded “Enterprise HRMS + HR Services Product Roadmap.” This document separates roadmap-derived direction from recommendations and assumptions introduced during discovery.

1. Product scope

1.1 MVP modules

Module	Proposed MVP capabilities
Core HRMS	Company and organization setup; departments, designations, locations and reporting managers; employee profiles and employment details; onboarding and employee status management; role-based access and basic audit history.
Leave & Attendance	Leave types and balances; leave requests and manager decisions; daily attendance records and working hours; attendance corrections/regularization; basic reports.
Basic Recruitment / ATS	Create job openings; store candidate profiles and resumes; track candidates through stages; record interview feedback and decisions; link successful candidates to employee onboarding.
Employee Self-Service	View and update permitted profile fields; submit and track leave requests; view attendance history; access authorized HR documents; basic notifications and request statuses.

1.2 Explicitly deferred from the initial MVP

Feature	Disposition
Payroll and statutory calculations	Later phase; requires separate compliance validation.
HR professional marketplace	Later phase.
Subscription billing implementation	Later phase; SaaS subscription remains the intended business model.
Performance, goals, learning and engagement	Later phase.
AI HR Copilot and workflow agents	Later phase.
Advanced SSO, enterprise integrations and analytics	Later phase.
Native mobile applications	Later phase; responsive web is the initial assumption.

1.3 Primary roles

Role	Initial responsibility
Super Admin	Platform-level tenant and governance management.
Company Admin	Company configuration, users and organization setup.
HR	Employee records, recruitment, leave, attendance and HR operations.
Manager	Authorized team visibility and approvals.
Employee	Self-service and access to own permitted HR information.

Authorization principle: a manager does not automatically receive access to every employee's sensitive data. Access should depend on role, tenant, relationship, and explicitly granted permissions.

2. Initial requirements baseline

These requirements are a discovery draft. Detailed user stories, validation rules, edge cases, and acceptance criteria are Phase 0.2 deliverables.

ID	Requirement	Priority
FR-01	A company can be registered and configured independently.	Must
FR-02	Authorized administrators can manage employee records.	Must
FR-03	Employees can authenticate securely and access permitted features.	Must
FR-04	Employees can submit leave requests and managers can decide them.	Must
FR-05	Authorized users can record and review attendance.	Must
FR-06	HR can create jobs and manage candidates through recruitment stages.	Must
FR-07	Users can view their permitted profile and HR information.	Must
FR-08	Sensitive changes and approval decisions are auditable.	Must
NFR-01	Tenant isolation prevents cross-company data access.	Critical
NFR-02	The application supports responsive desktop and mobile browsers.	Must
NFR-03	The architecture supports additional modules without major redesign.	Must
NFR-04	Backups, error logging and secure configuration are planned before launch.	Must

3. Assumptions and unresolved questions

Confirmed project constraints

- SMBs are the initial customer segment; the platform should retain enterprise scalability.
- India is the initial market.
- The project is developed by one developer with ChatGPT assistance.
- The first MVP includes Core HRMS, Leave & Attendance, Basic Recruitment, and Employee Self-Service.

Assumptions to validate

- Initial delivery is a browser-based SaaS application.
- Each company is represented as a tenant, with authorized users belonging to a company context.
- Email-based user accounts are acceptable for the MVP.
- Managers approve leave and relevant attendance corrections.
- Attendance is manual initially; no biometric or GPS integration is required.
- Recruitment is basic and not AI-driven in the first release.

Open decisions

- Identity provider and pricing tier.
- Authentication versus authorization boundaries and company membership model.
- ORM, database migration tooling, and whether PostgreSQL Row-Level Security will supplement application-level tenant checks.

- Attendance policies: missed check-outs, duplicate events, overnight shifts, timezone handling and correction rules.
- MVP success measures, initial customer/employee scale, and launch budget.

4. Architecture evaluation

The following preferences align with the uploaded roadmap and are provisionally accepted for planning. Specific vendor and implementation choices remain open where indicated.

Area	Preference	Evaluation / trade-off
Database	PostgreSQL	Good fit for relational HR records and transactional workflows. Local use is free; managed hosting, backups and operations cost money later.
Backend	NestJS + TypeScript	Matches the roadmap; supports modular APIs. Requires investment in TypeScript, Node.js and NestJS conventions.
Authentication	Managed identity provider	Reduces identity infrastructure operations; introduces provider pricing, quotas and vendor dependency. Candidate vendor not yet selected.
Attendance	Employee manual check-in/check-out	Low initial complexity. Requires server-side timestamps, duplicate handling, correction workflow and audit trail.
Deployment	Local first, then AWS	Controls early cloud spend. Containerization, environment configuration, migrations and deployment automation should be prepared early.

4.1 Recommended application structure

Frontend: Next.js + React + TypeScript, with Tailwind CSS and shadcn/ui as suggested by the roadmap.
Identity: external identity provider plus application-level permission and tenant membership checks.
Backend: NestJS modular monolith with separate modules for Core HRMS, Leave, Attendance, Recruitment, Self-Service, Authorization and Audit.
Data: PostgreSQL; tenant-owned records carry tenant context; migrations are version-controlled.
Files: private local development storage initially; AWS S3 is the roadmap-aligned production candidate.
Deployment: local development, then a test/staging AWS environment, then production after readiness checks.

4.2 Key architectural trade-offs

- **Modular monolith over microservices:** lower operational complexity for a solo developer; keep module boundaries clear so extraction remains possible if justified.
- **Tenant isolation:** enforce tenant scope in services and data access; evaluate PostgreSQL RLS as an additional safeguard, not as a substitute for authorization tests.
- **Managed identity:** reduces password and identity infrastructure burden, but company-level roles and permissions still belong to the application.
- **Local-first:** avoids early cloud spend; use Docker and production-like configuration so deployment is repeatable.
- **No premature infrastructure:** defer Kubernetes, microservices, Kafka, separate AI services and complex caching until actual needs justify them.

4.3 Security requirements

- Deny access by default on protected operations; validate user, tenant, role and resource permissions on the server.
- Use least-privilege database credentials, secure production connections, and secret management.
- Store HR documents privately and provide access only after authorization.
- Record actor, timestamp, tenant, entity and change details for sensitive actions.
- Plan backups, restoration testing, monitoring and secure deployment before production use.
- Do not treat browser-provided timestamps as authoritative attendance event times.

5. Cost and operational considerations

Decision	Initial impact	Later consideration
PostgreSQL	Free locally	Managed database, backups and operations.
NestJS	Framework is open source; developer time is the main initial investment	Maintenance and learning costs.
Managed identity	May fit a free allowance depending on provider and usage	Pricing tiers, quotas, provider dependency and migration effort.
Manual attendance	Low initial build complexity	Policy exceptions, corrections and customer-specific needs.
Local-first development	Low initial infrastructure spend	Deployment and operations work must be budgeted.
AWS	Costs begin when billable resources are provisioned	Compute, database, storage, networking, logs, backups and monitoring.

A credible monthly budget cannot yet be calculated because expected tenant count, employee count, storage, availability targets, region and identity-provider pricing are not defined. Avoid provisioning paid cloud resources until the initial environment and budget are agreed.

6. MVP success criteria — proposed for confirmation

- A company administrator can set up a tenant and its organization structure.
- Authorized HR users can create and maintain employee records.
- Employees can submit leave requests and managers can approve or reject them.
- Employees can record attendance and review their attendance history.
- HR can create a job opening and move a candidate through defined recruitment stages.
- Employees can access their own permitted information through self-service.
- Automated tests demonstrate that one tenant cannot access another tenant's data.
- Sensitive changes and approvals generate audit events.

These are proposed outcome checks, not measured performance commitments. Quantitative targets and acceptance criteria will be defined in Phase 0.2.

7. Phase 0.1 decision register

Status	Decision
Accepted for planning	PostgreSQL; NestJS + TypeScript; manual employee check-in/check-out; local development first, then AWS.
Conditionally accepted	Managed identity provider, pending vendor/pricing/capability review.
Recommended	Next.js frontend; modular monolith; API-first design; tenant-aware authorization; private file storage; Docker-based local environment.
Not yet locked	Identity vendor, ORM, RLS policy, AWS services, cloud budget, detailed attendance rules and measurable launch targets.

8. Phase 0.1 exit checklist

Exit criterion	Status
Target customer and initial market defined	Complete
MVP modules and deferred features identified	Complete
Initial roles and requirements drafted	Complete
Preferred stack evaluated	Complete — provisional
Identity provider and supporting implementation choices selected	Open
MVP success criteria agreed	Open

9. Next phase

Phase 0.2 — Software Requirements Specification (SRS): produce detailed user stories, workflows, acceptance criteria, role-permission matrix, validation rules, edge cases, and a testable definition of done. Do not begin full module implementation until requirements are sufficiently clear.

Reference source: uploaded “Enterprise HRMS + HR Services Product Roadmap.” This Phase 0.1 document retains the roadmap's product direction while identifying MVP scope, assumptions, and provisional recommendations.